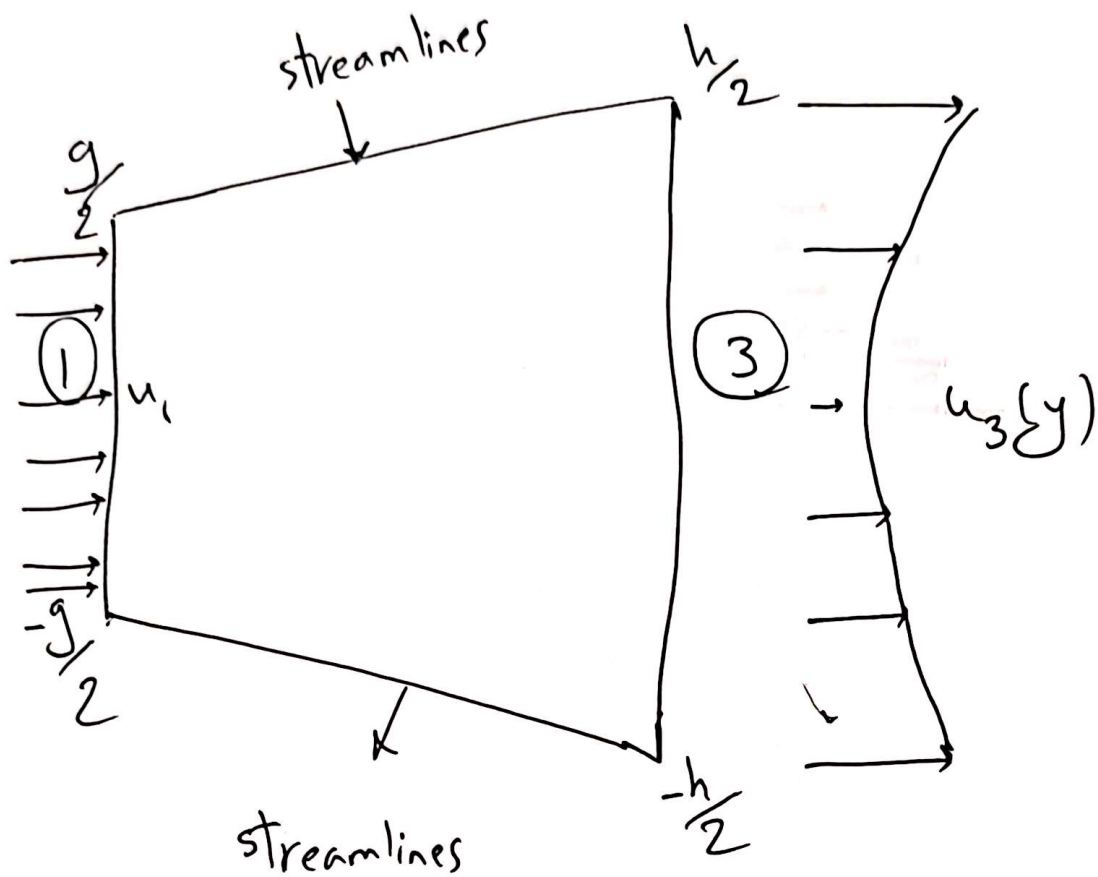


We have two sources of force $\begin{cases} P \\ \tau \end{cases}$

For airfoils, we assumed $P = P_\infty$ over all the surfaces, so drag is only due to momentum loss (due to viscosity) $\Rightarrow D = - \iint \tau(\vec{V} \cdot \vec{n}) u$

but for cylinders, we ignored viscosity (remember the definition of potential flow), so we have lift & drag just due to pressure imbalance in y & x directions. If for airfoils, you have different P on right & left, you should add pressure drag as well. i.e: $D_v = - \iint \tau(\vec{V} \cdot \vec{n}) u$

$$D_p = \iint_{A_{\text{exit}}} (P_{\text{exit}} - P_\infty) dA$$



$$\int_{-g/2}^{g/2} \rho u_1 dy = \int_{-h/2}^{h/2} \rho u_3(y) dy$$

For this case, as $P_3 = P_1$, drag is due to momentum loss, i.e. $\rightarrow D = - \oint \rho (\mathbf{V} \cdot d\mathbf{s}) u = F_x$
 airfoil $\rightarrow L = \oint \rho (\mathbf{V} \cdot d\mathbf{s}) v = F_y$

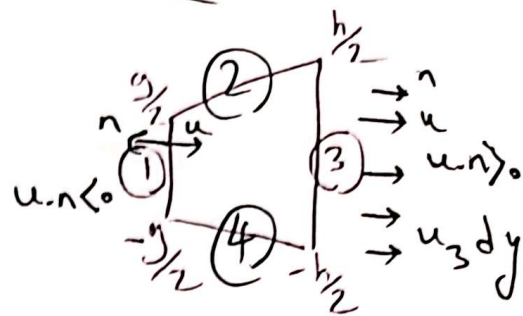
$$\text{cyl} \rightarrow L_{\text{cyl}} = \int (P_L - P_u) dy$$

inviscid flow so there is no loss in momentum due to viscosity

$$D = - \left[\int \rho \mathbf{V} \cdot d\mathbf{s} u \right]$$

$$D = - \left[\int_{-h/2}^{h/2} \rho V \cdot ds u - \int_{-g/2}^{g/2} \rho V \cdot ds u \right]$$

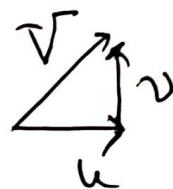
$$D = - \left[\int_{-h/2}^{h/2} \rho u_3^2 dy - \int_{-g/2}^{g/2} \rho u_1^2 dy \right]$$



Eq. (1) for the drag

Continuity $\left[\int_{-g/2}^{g/2} \rho u_1 dy = \int_{-h/2}^{h/2} \rho u_3 dy \right] \times u_1$

$$\left[\int_{-g/2}^{g/2} \rho u_1^2 dy = \int_{-h/2}^{h/2} \rho u_3 u_1 dy \right] \quad (2)$$



replace $\int_{-g/2}^{g/2} \rho u_1^2 dy$ from (2) in (1), we get:

$$D = - \left[\int_{-h/2}^{h/2} \rho u_3^2 dy - \int_{-h/2}^{h/2} \rho u_3 u_1 dy \right] = - \left[\int_{-h/2}^{h/2} \rho u_3 (u_3 - u_1) dy \right]$$

Eq. 3 for drag

multiply continuity by u_3 , you get:

$$\int_{-g/2}^{g/2} \rho u_1 u_3 dy = \int_{-h/2}^{h/2} \rho u_3^2 dy \quad (4)$$

replace $\int_{-h/2}^{h/2} \rho u_3^2 dy$ from (4) in (1), we get:

$$D = - \left[\int_{-g/2}^{g/2} \rho (u_1 u_3 - u_1^2) dy \right] = - \left[\int_{-g/2}^{g/2} \rho u_1 (u_3 - u_1) dy \right] \quad (5)$$

Eq. (1), (3) & (5) all give the same value for drag

Cartesian 2D $\rightarrow \nabla \cdot \vec{v}$

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2.11) Let us apply the integral form of the continuity equation. Note that, since surfaces (3) and (4) are streamlines, flow passes through only surfaces (1) and (2).

$$\frac{\partial}{\partial t} \iiint \rho d(\text{vol}) + \iint \rho \vec{V} \cdot \hat{n} dA = 0$$

Since the flow is incompressible and steady, we can write the continuity equation as

$$-\rho U_{\infty} \int_{-H_u}^{+H_u} dy + \rho U_{\infty} \int_{-H_D}^0 \left(-\frac{y}{H_D}\right) dy$$

$$\leftarrow \textcircled{1} \rightarrow \quad \leftarrow \textcircled{2a} \rightarrow$$

$$+ \rho U_{\infty} \int_0^{H_D} \left(\frac{y}{H_D}\right) dy = 0$$

$$\leftarrow \textcircled{2b} \rightarrow$$

(Refer to Problem 2.10 to see how to handle the v -component of velocity at station (2).)

$$-\rho U_{\infty} (2H_u) - \frac{\rho U_{\infty}}{H_D} \left(\frac{y^2}{2}\right) \Big|_{-H_D}^0 + \frac{\rho U_{\infty}}{H_D} \left(\frac{y^2}{2}\right) \Big|_0^{H_D} = 0$$

Rearranging and dividing through by ρU_{∞} (which is a common factor to every term), we obtain:

$$H_u = \frac{1}{2} H_D$$

2.12) Let us apply the integral form of the continuity equation.

Note that the effects of viscosity have caused a significant reduction of velocity in the wake of the airfoil (at station (2)).

As a result, there is a v -component of velocity which produces a mass flux across planes (3) and (4), because they are horizontal (perpendicular to the v -component).

Cartesian 2D $\rightarrow \nabla_x V = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$

2.28

This is very similar to Problem 2.27, except that the side boundaries of the control volume are streamlines. Thus, instead of using the continuity equation to determine the flow through sides ③ and ④ as was done for Problem 2.27, the continuity equation must be used to determine the relation between H_u and H_D .

Again, the pressure is constant over the external surface of the control volume for this steady, incompressible flow. Thus, the only force acting on the system of the fluid particles within the control volume is the negative of the drag.

$$-d = \oint_{-H_u}^{+H_u} [(U_\infty \hat{i}) \cdot (-\hat{i} dy)] U_\infty \quad D = - \iint (\mathbf{V} \cdot \mathbf{n}) u$$

← ① →

$$+ \oint_{-H_D}^0 \left[\left(-U_\infty \frac{y}{H_D} \hat{i} - \cancel{U_\infty \hat{j}} \right) \cdot (\hat{i} dy) \right] \left(-U_\infty \frac{y}{H_D} \right)$$

← ② →

$$+ \oint_0^{H_D} \left[\left(U_\infty \frac{y}{H_D} \hat{i} + \cancel{U_\infty \hat{j}} \right) \cdot (\hat{i} dy) \right] \left(U_\infty \frac{y}{H_D} \right)$$

→ ②

$$\begin{aligned} & (\mathbf{V} \cdot \mathbf{n}) u \\ & \downarrow \\ & u = U_\infty \left(-\frac{y}{H_D} \right) \\ & \mathbf{V} = \left(-U_\infty \frac{y}{H_D} \right) \hat{i} \\ & \quad - U_\infty \hat{j} \end{aligned}$$

There is no momentum transport across boundaries ③ and ④, since they are streamlines.

$$-d = \oint U_\infty^2 [2H_u] + \oint U_\infty^2 \left[\frac{y^3}{3H_D^2} \right]_{-H_D}^0 + \oint U_\infty^2 \left[\frac{y^3}{3H_D^2} \right]_0^{+H_D}$$

$$d = \oint U_\infty^2 \left[2H_u - \frac{2}{3} H_D \right]$$

$$= -U_\infty^2 \frac{y}{H_D}$$

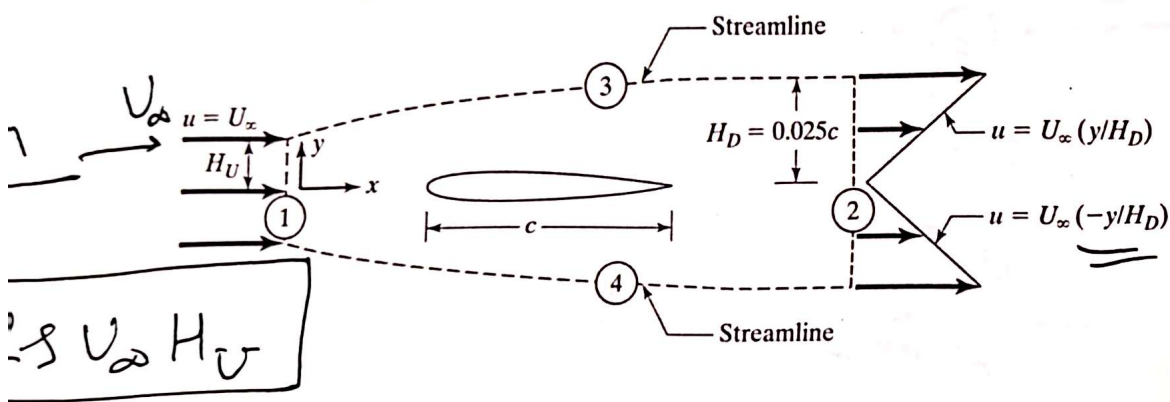
Cartesian 2D $\rightarrow \nabla_x V = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$

1. Velocity profiles are measured at the upstream end (surface 1) and at the downstream end (surface 2) of the control volume shown below. The flow is incompressible, two-dimensional and steady. The surfaces 3 and 4 are streamlines. The dimension $H_D = 0.025c$ where c is the chord of the airfoil. The pressure is p_∞ (a constant) over the entire surface of the airfoil.

(a) What is the vertical dimension for the upstream station H_U ?

(b) What is the drag coefficient $C_d = d / (1/2 \rho_\infty U_\infty^2 c)$ of the airfoil?

$$C_d = \frac{d}{\frac{1}{2} \rho U_\infty^2 c}$$



2.28 Contd.] We can use the integral continuity equation to determine the relation between H_U and H_D for this steady, incompressible flow;

$$+ \int_{-H_U}^{+H_U} [(U_\infty \hat{i}) \cdot (-\hat{i} dy)] + \int_{-H_D}^0 \left[\left(-\frac{U_\infty y}{H_D} \hat{i} - v \hat{j} \right) \cdot (\hat{i} dy) \right]$$

$\xleftarrow{\quad} \textcircled{1} \xrightarrow{\quad} \quad \quad \quad \xleftarrow{\quad} \textcircled{2} \xrightarrow{\quad}$

$$+ \int_0^{H_D} \left[\left(+\frac{U_\infty y}{H_D} \hat{i} + v \hat{j} \right) \cdot (\hat{i} dy) \right] = 0$$

$\xleftarrow{\quad} \textcircled{2} \xrightarrow{\quad}$

Thus,
$$U_\infty 2H_U - \frac{U_\infty y^2}{2H_D} \Big|_{-H_D}^0 + \frac{U_\infty y^2}{2H_D} \Big|_0^{H_D} = 0$$

Therefore,
$$U_\infty 2H_U = U_\infty H_D$$

as was shown in Problem 2.11, $H_U = \frac{1}{2} H_D$

$$d = \rho U_\infty^2 \left[H_D - \frac{2}{3} H_D \right] = \frac{1}{3} \rho U_\infty^2 H_D$$

$$C_d = \frac{d}{\frac{1}{2} \rho U_\infty^2 c} = \frac{\frac{1}{3} \rho U_\infty^2 \left(\frac{1}{40} c \right)}{\frac{1}{2} \rho U_\infty^2 c} = \frac{1}{60} = 0.0167$$

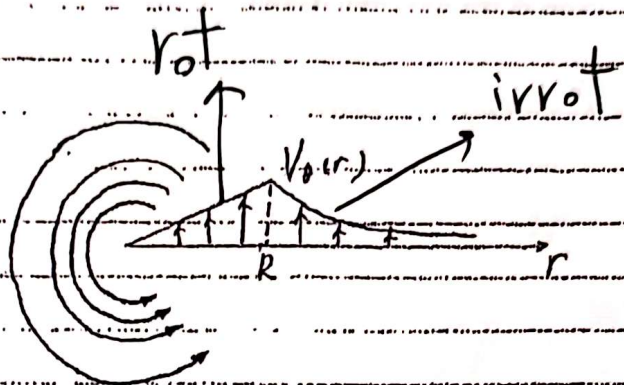
As one would expect, we have gotten the same result as was obtained in Problem 2.28. Therefore, the result is not dependent on the control volume.

Cartesian 2D $\rightarrow \nabla \times \vec{v} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$

Problem #2

$V_r = 0$

$V_\theta = \begin{cases} \omega r & \text{for } r < R \\ \omega R^2 / r & \text{for } r > R \end{cases}$



a) In cylindrical coordinates

Cyl. $\begin{cases} V_r \\ V_\theta \\ V_z \end{cases}$ 3-D $\nabla \times \vec{v} = \left(\frac{1}{r} \frac{\partial V_z}{\partial \theta} - \frac{\partial V_\theta}{\partial z} \right) \vec{e}_r + \left(\frac{\partial V_r}{\partial z} - \frac{\partial V_z}{\partial r} \right) \vec{e}_\theta + \left(\frac{1}{r} \frac{\partial (r V_\theta)}{\partial r} - \frac{\partial V_r}{\partial \theta} \right) \vec{e}_z$ For 2-D

For 2-D flow $\nabla \times \vec{v}$ only has z component.

For $r < R$, $(\nabla \times \vec{v})_z = \frac{1}{r} \frac{\partial (\omega r^2)}{\partial r} = 2\omega \neq 0$, rotational

For $r > R$, $(\nabla \times \vec{v})_z = \frac{1}{r} \frac{\partial \omega^2}{\partial r} = 0$, irrotational

b) r-momentum eqn. is

$V_r \frac{\partial V_r}{\partial r} + \frac{V_\theta}{r} \frac{\partial V_r}{\partial \theta} - \frac{V_\theta^2}{r} = -\frac{1}{\rho} \frac{\partial p}{\partial r}$

Since $V_r = 0$, we have

$-\frac{1}{\rho} \frac{\partial p}{\partial r} = -\frac{V_\theta^2}{r}$

$\Rightarrow \frac{\partial p}{\partial r} = \rho V_\theta^2 \cdot \frac{1}{r}$

Ans. can find $\frac{\partial p}{\partial \theta} = 0$ from θ -momentum eqn. thus $p = p(r)$

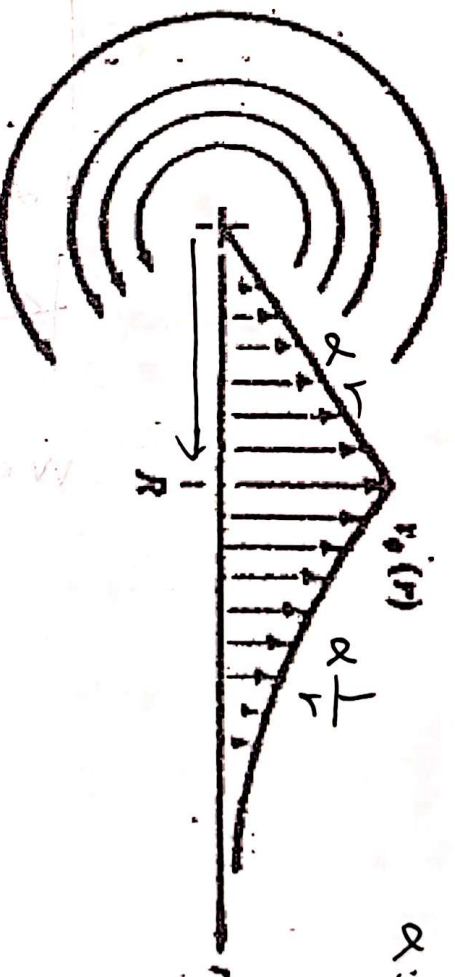
$\therefore p = \int \frac{\rho V_\theta^2}{r} dr$

Define pressure as p_1 for $r < R$, p_2 for $r > R$

Boundary condition: $r \rightarrow \infty$, $p_2 = p_\infty$
 $r = R$, $p_1(R) = p_2(R)$

A hurricane may be modeled as a Rankine vortex as shown below, with $v_r=0$ and $v_\theta(r)$ such that:

$$v_\theta = \begin{cases} \omega r & \text{for } r < R \\ \omega R^2/r & \text{for } r > R \end{cases}$$



a) Determine whether this flow pattern is irrotational in either the inner or the outer region.

b) Determine the pressure distribution $p(r)$ in the tornado, assuming $p=p_\infty$ as $r \rightarrow \infty$, and the location and the magnitude of the lowest pressure.

c) Assume that the core has a radius of 10 miles, and maximum rotational speed 185 mph. Find the minimum pressure. Is this similar to values recorded for Irma?

Hint: The r-momentum equation (steady, no body forces, incompressible, inviscid):

$$V_r \frac{\partial V_r}{\partial r} + \frac{V_\theta}{r} \frac{\partial V_r}{\partial \theta} = -\frac{V_\theta^2}{r} = -\frac{1}{\rho} \frac{\partial p}{\partial r}$$

for $r > R$,
 $0 \Rightarrow p_2(r) = p \int \frac{\omega^2 R^4}{r^3} dr$

$$= p \omega^2 R^4 \left(-\frac{1}{2r^2} \right) + C_1$$

Applying Boundary condition, $r \rightarrow \infty$, $p_2 = p_\infty$

We can get $C_1 = p_\infty$

$$\therefore p_2(r) = -\frac{1}{2} \frac{\omega^2 R^4}{r^2} + p_\infty$$

for $r < R$,

$$0 \Rightarrow p_1(r) = p \int \omega^2 r dr$$

$$= \frac{1}{2} p \omega^2 r^2 + C_2$$

To determine C_2 applying the second B.C.

$$p_1(R) = p_2(R) \text{ gives}$$

$$\frac{1}{2} p \omega^2 R^2 + C_2 = -\frac{1}{2} p \omega^2 R^2 + p_\infty$$

$$\therefore C_2 = p_\infty - p \omega^2 R^2$$

$$\therefore p_1(r) = \frac{1}{2} p \omega^2 r^2 + p_\infty - p \omega^2 R^2$$

Obviously, the lowest pressure takes place at $r=0$, where $p_1(0) = p_\infty - p \omega^2 R^2$

* $V_0 = V_0(r)$

0 - Momentum:

$$\frac{1}{r} \frac{\partial V_0}{\partial r} + \frac{V_0}{r} \frac{\partial V_0}{\partial \theta} + \frac{V_0}{r} \frac{\partial V_0}{\partial z} = -\frac{1}{\rho r} \frac{\partial p}{\partial \theta}$$

$$\Rightarrow \frac{\partial p}{\partial \theta} = 0 \Rightarrow p = p(r) //$$

(185 mph)

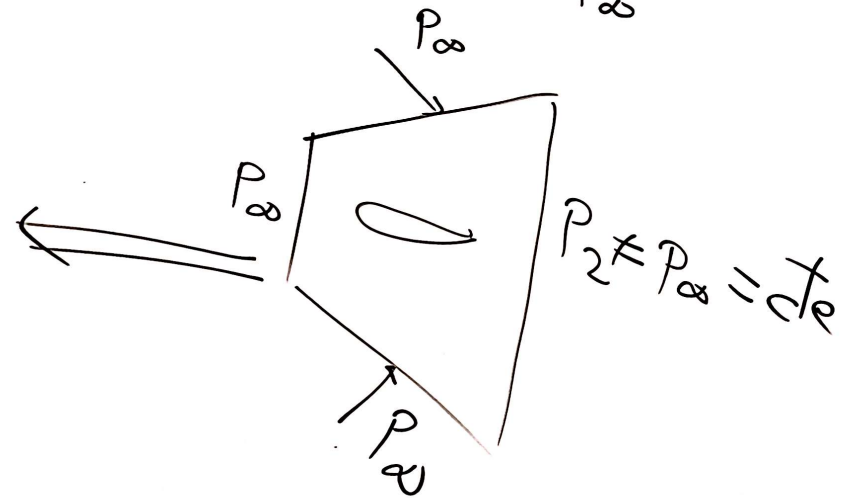
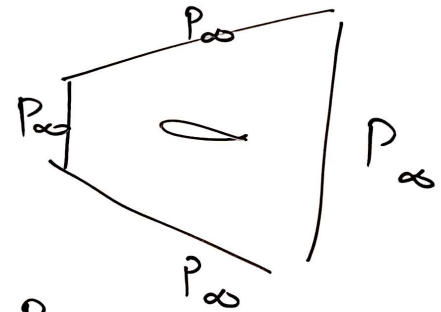
Irma $p_{\text{lowest}} = 101,325 - 1.225 \frac{\text{kg}}{\text{m}^3} \times \left(\frac{82.25 \text{ m}}{\text{s}} \right)^2 = 92,981$

lowest recorded 914 mbars)

930 mbars

$D = ?$ Momentum lost + Pressure difference
 $L = ?$ viscous drag

Zero in



if $P_2 = P_2(y)$
 Add $\int_{-h/2}^{h/2} (P_2 - P_\infty) dy$
 to viscous drag

Add $(P_2 - P_\infty) A_2$
 to drag